

Short Answer Test

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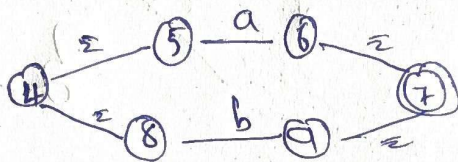
1). Given $E \rightarrow E + E / id$
 $E \rightarrow E + E$
 $E' \rightarrow id E'$
 $E' \rightarrow + E E' / \Sigma$

It is ambiguous.

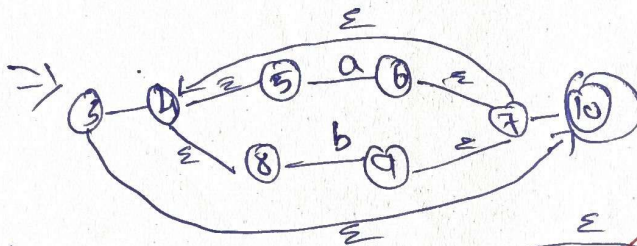
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70
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2). $a(a/b)^*a$

$a/b \Rightarrow$

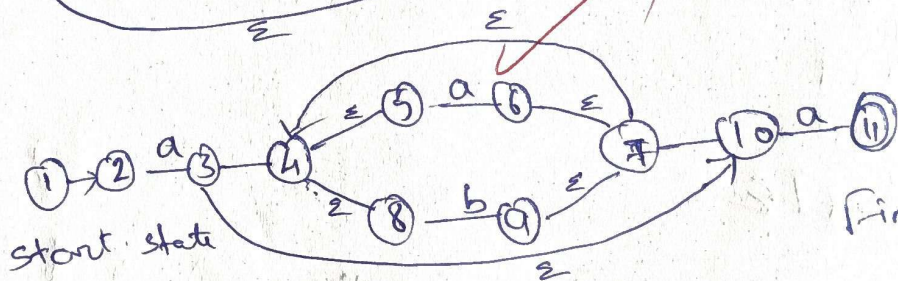


$(a/b)^* \Rightarrow$



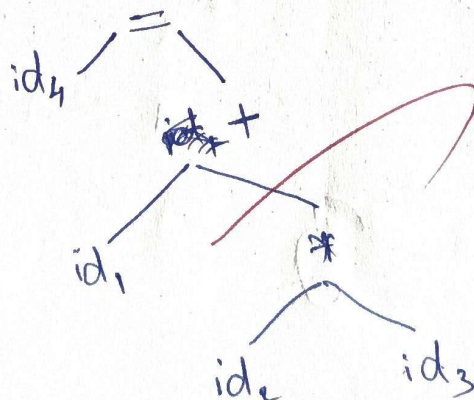
$a(a/b)^*a$

$\Rightarrow$



3). $id + id * id = S$

$\langle id_1 \rangle \langle + \rangle \langle id_2 \rangle \langle * \rangle \langle id_3 \rangle = \langle id_4 \rangle$



$$\begin{aligned}
 4. \quad & A \rightarrow A\alpha/\beta \\
 & A \rightarrow A\alpha \\
 & A \rightarrow \beta \\
 & A' \rightarrow \beta A' \\
 & A' \rightarrow \alpha A'/\Sigma
 \end{aligned}$$

$$\begin{aligned}
 5. \quad & S \rightarrow iEtS/iEtSeS/a \\
 & A \rightarrow \alpha B/\alpha \vee \\
 & A' \rightarrow B/\vee \\
 \Rightarrow & S' \rightarrow iEtS \\
 & S' \rightarrow es/\Sigma
 \end{aligned}$$

$$\begin{aligned}
 \text{Here } \alpha &= iEtS \\
 \beta &= - \\
 \vee &= es
 \end{aligned}$$

$$\begin{aligned}
 S &\rightarrow iEtS \\
 S &\rightarrow iEtSs' \\
 S &\rightarrow iEtS/iEtSs'/\Sigma
 \end{aligned}$$

$$\begin{aligned}
 6. \quad & S \rightarrow aBph \\
 & B \rightarrow cL \\
 & C \rightarrow bC/\Sigma \\
 & D \rightarrow EF \\
 & E \rightarrow g/\Sigma \\
 & F \rightarrow f/\Sigma
 \end{aligned}$$

$$\begin{aligned}
 \text{First}(S) &: \{a\} \\
 \text{First}(B) &: \{c\} \\
 \text{First}(C) &: \{b, \Sigma\} \\
 \text{First}(D) &: \{g, f, \Sigma\} \\
 \text{First}(E) &: \{g, \Sigma\} \\
 \text{First}(F) &: \{f, \Sigma\}
 \end{aligned}$$

$$\begin{aligned}
 \text{Follow}(S) &: \{\$ \} \\
 " \quad B &: \{g, f, \$ \} \\
 " \quad C &: \{g, f, \$ \} \\
 " \quad D &: \{g, f, \$ \} \\
 " \quad E &: \{g, f, \$ \} \\
 " \quad F &: \{g, f, \$ \}
 \end{aligned}$$

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$$1. S \rightarrow sa|sb|cd.$$

$$A \rightarrow \alpha B | \alpha V$$

$$A' \rightarrow \alpha / \beta$$

$$S' \rightarrow \alpha / \beta / \Sigma$$

$$S \rightarrow SS' / SS' / \Sigma$$

$$8. E \rightarrow TE'$$

$$E' \rightarrow +TE' / \Sigma$$

$$\text{First}(E') : \{+, \Sigma\}$$

$$9. S \rightarrow (S)S / \Sigma$$

$$S(0) \Rightarrow S \rightarrow (0)S / \Sigma$$

$$\Rightarrow S \rightarrow 0 / \Sigma$$

"0 / Σ " which is not valid.

1d) Rules to follow in shift reduce parsing:

1. Start with dollar "\$".

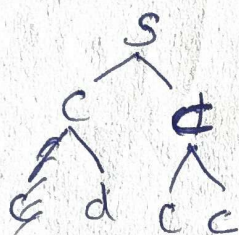
2. The order (or) precedence of operators is, $(id > () > * > + > E \> \$)$

3. If we take the terminal which is greater than the present, then shift the terminal.

4. Else reduce to the non-terminal given in question.

5. Repeat the process until "\$" remains at last.

11). Grammar $S' \rightarrow s$
 $s \rightarrow cc$
 $c \rightarrow cc \mid d$
 item: $[s' \rightarrow \cdot s]$



12). $S \rightarrow a sb / ab$

$S \rightarrow AB$

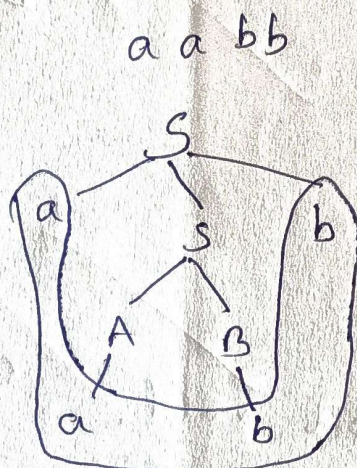
$A \rightarrow a$

$B \rightarrow b$

First (S) : {a, b}

First (A) : {a}

First (B) : {b}



(Parse found.)

13). $S \rightarrow A$
 $A \rightarrow a B \mid A d$
 $B \rightarrow b$
 $C \rightarrow g$

$S \rightarrow A$

$A \rightarrow a B A'$

$A' \rightarrow d A' \mid a B A' d A'$

$A' \rightarrow a B A' / \epsilon$

$S \rightarrow A$

$\Rightarrow A \rightarrow a B A'$

$A' \rightarrow d A'$

$B \rightarrow b$

$C \rightarrow g$

First (S) : {a}

First (A) : {a}

First (A') : {d}

First (B) : {b}

First (C) : {g}

Follow(S) : { }

Follow(A) : { }

Follow(A') : { }

Follow(B) : {d, }

Follow(C) : {a, }

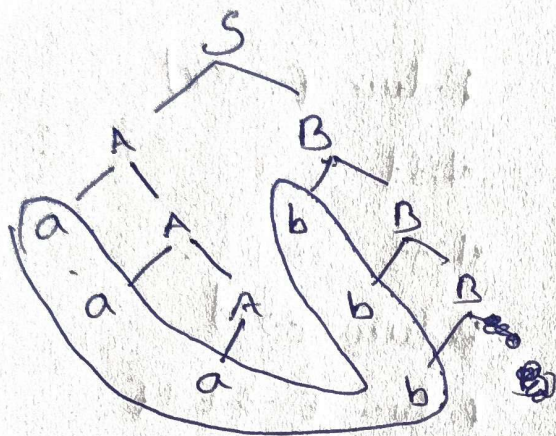
$$S \rightarrow AB$$

$$A \rightarrow aA/a$$

$$B \rightarrow bB/b$$

⑤

aaa bbb



Hence found.

16. $S \rightarrow AB$

$$A \rightarrow aA/\epsilon$$

$$B \rightarrow bB/c$$

First (S): {a, b, c, ϵ }

First (A): {a, ϵ }

First (B): {b, c}

Follow (S): { ϵ }

Follow (A): {b, c}

Follow (B): {b, c, ϵ }

17. 1. Check for the non-terminal ϵ in production at the first place. It must be equal to the input.

2. If No - non-terminal at first place, then it is in stable (or) optimized state.

3. Else reduce it, by

$$A \rightarrow \alpha \beta / \alpha \beta$$

$$A \rightarrow \alpha A'$$

$$A' \rightarrow \beta / \beta$$

$$A \rightarrow A\alpha / \beta$$

$$A \rightarrow \beta A'$$

$$A' \rightarrow \alpha A' / \epsilon$$

18. $id + id * id$

<u>Shift</u>	<u>reduce</u>	<u>Parsing</u> <u>ILP built</u>
\$		$id + id * id \$$
\$ id		$+ id * id \$$
\$ E		$+ id * id \$$
\$ E +		$id * id \$$
\$ E + id		$* id \$$
\$ E + E		$* id \$$
\$ E		$id \$$
\$ E *		\$
\$ E * id		\$
\$ E * E		\$
\$ E		

<u>Recursive</u> <u>Stack</u>	<u>descent</u>	<u>Parsing</u> <u>ILP built</u>
\$		$id + id * id \$$
\$ E		$id + id * id \$$
\$ E' T		$id + id * id \$$
\$ E' T' F		$id + id * id \$$
\$ E' T' id		$id + id * id \$$
\$ E'		$+ id * id \$$
\$ E' T *		$* id * id \$$
\$ E' T' F		$id * id \$$
\$ E' T' id		$id * id \$$

$E \rightarrow id$
 $E \rightarrow E + E$
 $E \rightarrow E * E$

<u>operation</u>
shift
Reduce $E \rightarrow id$
shift
shift
Reduce $E \rightarrow id$
Reduce $E \rightarrow E + E$
shift
shift
reduce $E \rightarrow id$
Reduce $E \rightarrow E * E$

<u>operation</u>
-
$E \rightarrow TE'$
$T \rightarrow FT'$
$F \rightarrow id$
$T' \rightarrow \epsilon$
$E' \rightarrow +TE'$
$T \rightarrow FT'$
$F \rightarrow id$
$E' \rightarrow \epsilon$

$\$id$

$\$$

$*id \$$

$id \$$

$\$$

$F \rightarrow id$

$T' \rightarrow \epsilon$

~~10~~ ~~11~~

$E \rightarrow TE'$

$E' \rightarrow +TE' / \epsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' / \epsilon$

$F \rightarrow (E) / id$

First(E): { (, id }

" (E'): { +, ϵ }

" (T): { (, id }

" (T'): { *, ϵ }

" (F): { (, id }

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Follow(E): { \$,) }

" E': { \$,) }

" T: { +, \$,) }

" T': { +, \$,) }

" F: { *, +, \$,) }

Table

Non-Terminal	id	(	)	+	*	\$
E	$E \rightarrow TE'$	$E \rightarrow TE'$				
E'			$E' \rightarrow \epsilon$	$E' \rightarrow +TE'$		$E' \rightarrow \epsilon$
T	$T \rightarrow FT'$	$T \rightarrow FT'$				
T'			$T' \rightarrow \epsilon$	$T' \rightarrow \epsilon$	$T' \rightarrow *FT'$	$T' \rightarrow \epsilon$
F	$F \rightarrow (E)$	$F \rightarrow (E)$				

1d). stack

\$
\$ id₁
\$ E
\$ E * +
\$ E + id₂
\$ E + E
\$ E
\$ E *
\$ E * id₃
\$ E * E
\$ E

I/O buffer

id₁ + id₂ * id₃ \$
+ id₂ * id₃ \$
+ id₂ * id₃ \$
id₂ * id₃ \$
* id₃ \$
* id₃ \$
* id₃ \$
id₃ \$
\$
\$
\$

operation

shift
reduce E → id₁
shift
shift
reduce E → id₂
reduce E → E + E
shift
shift
reduce E → id₃
reduce E → E * E

2d).
int a = 10, b = 20;
int sum = a + b;
printf("%d", sum);

(a). The phases of compiler are:

1. Lexical analyzer:

Token: a, b, sum, =, 10, 20, ;

2. Syntax analyzer:

It is also known as parse tree.

sum = a + b or sum = 10 + 20.

3. Semantic Analyzer:

It checks the semantic errors.

4. Intermediate code generation:

It gives intermediate code.

5. Code Analyzer

It analyzes the code.

6. Code optimizer:

It optimizes the code.

$\langle id_1 \rangle \langle + \rangle \langle id_3 \rangle$

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